Example 11.2

Integrate the following with respect to x.

(i) $\frac{1}{\cos^2 x}$ (ii) $\frac{\cot x}{\sin x}$ (iii) $\frac{\sin x}{\cos^2 x}$ (iv) $\frac{1}{\sqrt{1-x^2}}$

Solution

(i) $\int \frac{1}{\cos^2 x} dx = \int \sec^2 x \, dx = \tan x + c$

(ii) $\int \frac{\cot x}{\sin x} dx = \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$

(iii) $\int \frac{\sin x}{\cos^2 x} dx = \int \frac{\sin x}{\cos x} . \frac{1}{\cos x} dx = \int \tan x \sec x \, dx = \sec x + c$

(iv) $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$

Example 11.3

Integrate the following with respect to x:

(i) $\frac{1}{e^{-x}}$ (ii) $\frac{x^2}{x^3}$ (iii) $\frac{1}{x^3}$ (iv) $\frac{1}{1+x^2}$

Solution

(i) $\int \frac{1}{e^{-x}} dx = \int e^x dx = e^x + c$

(ii) $\int \frac{x^2}{x^3} dx = \int \frac{1}{x} dx = \log|x| + c$

(iii) $\int \frac{1}{x^3} dx = \int x^{-3} dx = \frac{x^{-3+1}}{-3+1} + c = -\frac{1}{2x^2} + c$

(iv) $\int \frac{1}{1+x^2} dx = \tan^{-1} x + c$

EXERCISE 11.1

Integrate the following with respect to x:

(1) (i) x^{11} (ii) $\frac{1}{x^7}$ (iii) $\sqrt[3]{x^4}$ (iv) $(x^5)^{\frac{1}{8}}$

(2) (i) $\frac{1}{\sin^2 x}$ (ii) $\frac{\tan x}{\cos x}$ (iii) $\frac{\cos x}{\sin^2 x}$ (iv) $\frac{1}{\cos^2 x}$

(3) (i) 12^3 (ii) $\frac{x^{24}}{x^{25}}$ (iii) e^x

(4) (i) $(1+x^2)^{-1}$ (ii) $\left(1-x^2\right)^{-\frac{1}{2}}$

11.4 Integrals of the Form $\int f(ax+b)dx$

We know that

$$\frac{d}{dx}\left[\frac{(x-a)^{10}}{10}\right] = (x-a)^9 \Rightarrow \int (x-a)^9 dx = \frac{(x-a)^{10}}{10} + c$$

$$\frac{d}{dx}[\sin(x+k)] = \cos(x+k) \Rightarrow \int \cos(x+k)dx = \sin(x+k) + c$$

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